

1. We have

$$M_t = \frac{1}{t} \int_0^t S_u du, \quad U_t = \frac{tM_t + (T-t)S_t}{T}.$$

Since S is continuous, by the continuity of the integral average

$$M_0 := \lim_{t \downarrow 0} M_t = \lim_{t \downarrow 0} \frac{1}{t} \int_0^t S_u du = S_0.$$

Thus

$$U_0 = \frac{0 \cdot M_0 + (T-0)S_0}{T} = S_0.$$

At maturity,

$$U_T = \frac{TM_T + (T-T)S_T}{T} = M_T.$$

So indeed $U_0 = S_0$ and $U_T = M_T$.

2. Write

$$A_t := \int_0^t S_u du, \quad U_t = \frac{A_t + (T-t)S_t}{T}.$$

Then $dA_t = S_t dt$ and, using that T is constant,

$$d((T-t)S_t) = -S_t dt + (T-t) dS_t.$$

Therefore

$$dU_t = \frac{1}{T} (dA_t + d((T-t)S_t)) = \frac{1}{T} (S_t dt - S_t dt + (T-t) dS_t) = \frac{T-t}{T} dS_t.$$

Over the infinitesimal interval $[t, t+dt]$ a self-financing portfolio with Δ_t units of S has dynamics $dV_t = \Delta_t dS_t$ (with $r = 0$). To have $V_t = U_t$ we must match $dV_t = dU_t$, hence

$$\Delta_t = \frac{T-t}{T}.$$

So U_t can indeed be replicated, by holding $\frac{T-t}{T}$ units of the stock and the rest in the money market account (here with zero rate).

3. From the previous computation and the stock dynamics

$$dS_t = \sigma(t)S_t d\widetilde{W}_t,$$

we obtain

$$dU_t = \frac{T-t}{T} dS_t = \frac{T-t}{T} \sigma(t)S_t d\widetilde{W}_t.$$

Equivalently,

$$dU_t = \sigma(t) \frac{T-t}{T} \frac{S_t}{U_t} U_t d\widetilde{W}_t.$$

This is the SDE satisfied by U_t .

4. From (3),

$$dU_t = \sigma(t) \frac{T-t}{T} \frac{S_t}{U_t} U_t d\widetilde{W}_t.$$

Assume that

$$\frac{S_t}{U_t} = 1 \quad \text{for all } t,$$

so that the SDE simplifies to

$$dU_t = \sigma_U(t) U_t d\widetilde{W}_t, \quad \sigma_U(t) := \sigma(t) \frac{T-t}{T}.$$

Thus U_t is a (driftless) geometric Brownian motion with deterministic volatility $\sigma_U(t)$. Hence

$$\ln U_T \sim \mathcal{N}\left(\ln U_0 - \frac{1}{2}v^2, v^2\right), \quad v^2 := \int_0^T \sigma_U^2(t) dt = \int_0^T \sigma^2(t) \left(\frac{T-t}{T}\right)^2 dt.$$

The Asian call payoff is $(M_T - K)^+ = (U_T - K)^+$. Under the risk-neutral measure with $r = 0$, its time-0 price is the Black-Scholes price of a call on U :

$$C_0^{\text{Asian}} = U_0 N(d_1) - K N(d_2),$$

where $U_0 = S_0$ from part (1),

$$d_1 = \frac{\ln\left(\frac{U_0}{K}\right) + \frac{1}{2}v^2}{v}, \quad d_2 = d_1 - v,$$

and $N(\cdot)$ is the standard normal cdf.

For a vanilla European call on S with the same S_0, K, T and volatility $\sigma(t)$, the total variance is

$$V^2 := \int_0^T \sigma^2(t) dt,$$

and the price (with $r = 0$) is

$$C_0^{\text{vanilla}} = S_0 N(D_1) - K N(D_2), \quad D_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \frac{1}{2}V^2}{V}, \quad D_2 = D_1 - V.$$

Since $\left(\frac{T-t}{T}\right)^2 \leq 1$ for all t , we have $v^2 \leq V^2$ and hence $v \leq V$. Because a call option price is increasing in the volatility, it follows that

$$C_0^{\text{Asian}} \leq C_0^{\text{vanilla}}.$$

In the special case of constant volatility $\sigma(t) \equiv \sigma$, we get

$$v^2 = \sigma^2 \int_0^T \left(\frac{T-t}{T}\right)^2 dt = \sigma^2 \frac{T}{3},$$

so the Asian call price equals the Black-Scholes price of a call on S_0 with the same K, T but with an “effective” volatility $\hat{\sigma} = \sigma/\sqrt{3}$, which is lower than σ . Consequently, the Asian call is cheaper than the corresponding vanilla European call.